Maths Review for Machine Learning and Data Science
Topic 2: Linear Algebra

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Introduction

Maths Review for Machine Learning and Data Science:

1. Functions and Notation
2. **Linear Algebra**
   - Vectors
   - Matrices
   - Converting equations to vector and matrix form
3. Calculus
   - Differentiation
   - Partial derivatives
   - Matrix calculus
4. Optimisation
   - Minima and maxima
   - Argmin and Argmax
   - Convex functions

Introduction

A few important notes:

- These workshops will review the fundamentals of each topics. Suggestions for further reading are given at the end those wishing to review or learn more advanced content. Go
- I won't be covering matrix operations in Python during this session, but one of the linked resources is a great place to start if you are looking for that.
- If you have any questions during the workshop, please raise your hand and ask! If you would like to discuss any of these topics in more detail, one-on-one, there is information at the end about accessing our free drop-in and consultation services. Go

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Vectors

A *vector* is is essentially a list of values.

For example, we might have vector

$$v = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix},$$

such that $v_1 = 3$, $v_2 = 5$, and $v_3 = 2$.

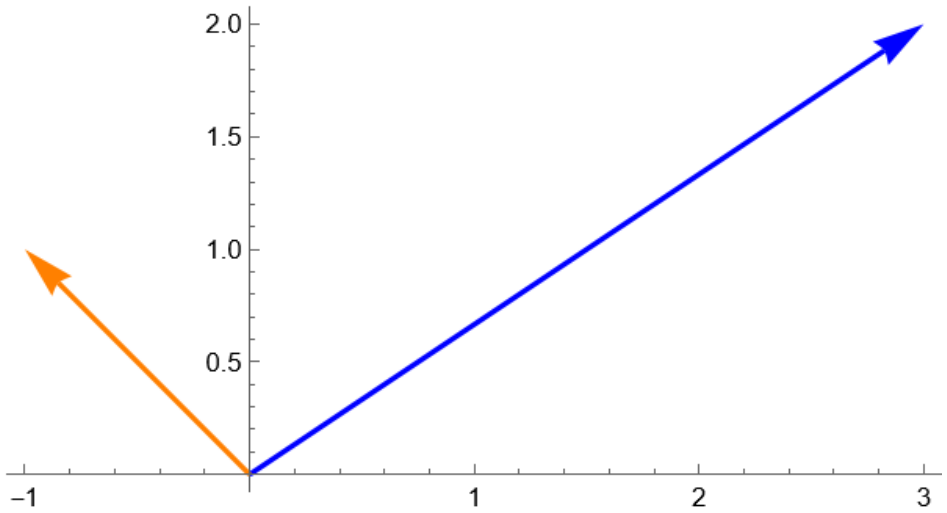
You might have noticed that I represented v as a vertical column, rather than a horizontal row. This is because we will assume from now on that all vectors are *column vectors*.

Even when they are written in a row, like this: $v = \begin{bmatrix} 3 & 5 & 2 \end{bmatrix}$, so that they fit nicely into a line of text.

Vectors

In physics, you might have seen vectors used to represent a line, or a direction and distance from the origin.

Here we have plotted the vectors $\begin{bmatrix} 3 & 2 \end{bmatrix}$ and $\begin{bmatrix} -1 & 1 \end{bmatrix}$.



This also applies when we move into 3 or more dimensions.

Vector Notation

To obtain a *row vector* we can take the *transpose*, $\top$, of a column vector:

$$v^\top = \begin{bmatrix} 3 & 5 & 2 \end{bmatrix}$$

We can use set notation to denote the *dimension* of a vector. That is,

$$x \in \mathbb{R}^n$$

tells us that x is a column vector with n rows.

Vector Notation

We use *subscript* to *index* a vector, so that x_i is the ith element of vector x, usually starting our indices at $i = 1$.

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

and

$$x^\top = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}.$$

Vector Addition

When we add two vectors together, we add corresponding elements. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

For example,

$$\begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \\ 4 \end{bmatrix}$$

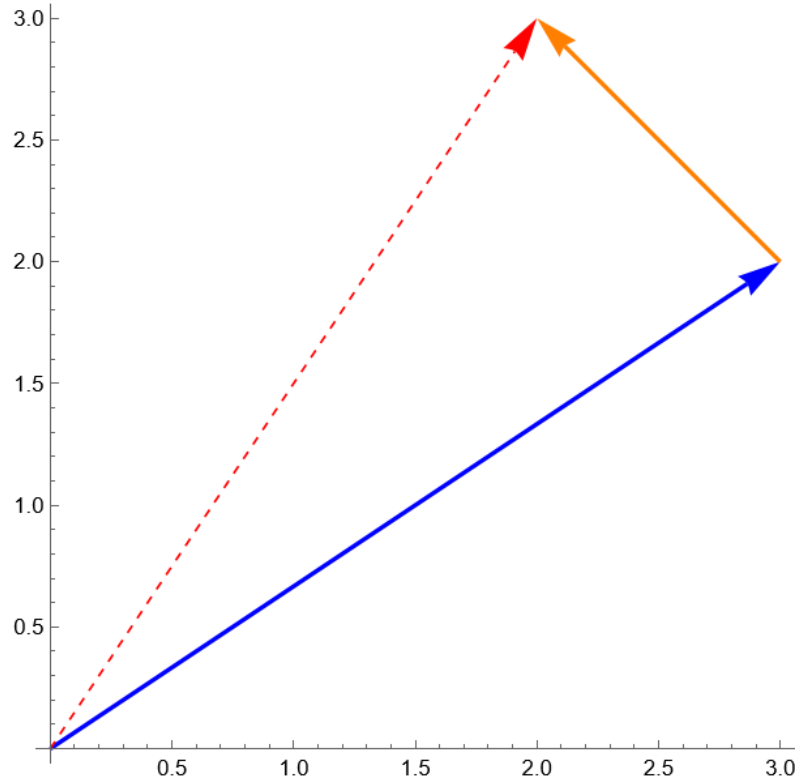
To perform vector addition, all vectors must have the same dimension.

Vector Addition

We can think of vector addition, geometrically, as adding two vectors end-to-end.

For example, the following plot shows:

$$\begin{bmatrix}3\\2\end{bmatrix} + \begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$$



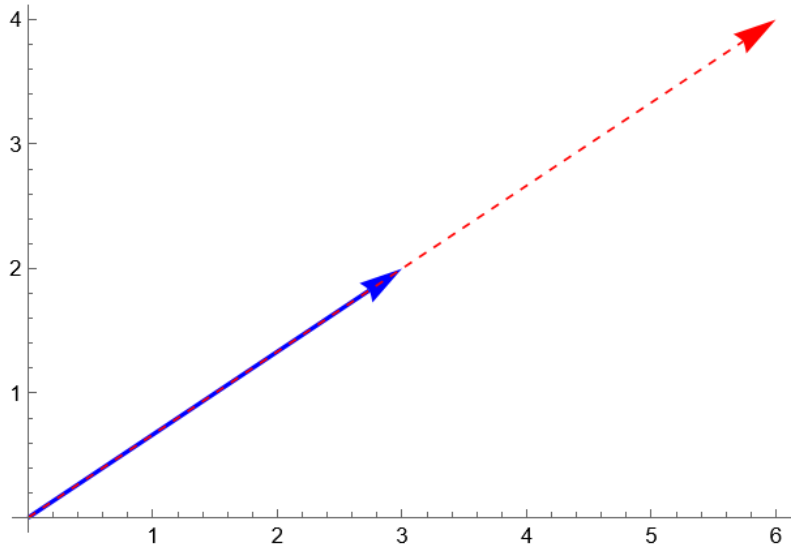
Vector Multiplication

To multiply a vector by a scalar value, simply multiply each element of the vector by the scalar.

$$c \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix}$$

This multiplies the magnitude of the vector, without affecting its direction. For example,

$$2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$



Vector Multiplication

We can multiply two vectors of the same dimension, by taking the *dot product*.

The dot product is defined as the sum of the products of corresponding elements of the vectors. That is,

$$u \cdot v = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

For example,

$$\begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix} = 3 \cdot 4 + 5 \cdot 7 + 3 \cdot 1 = 12 + 35 + 3 = 50$$

The dot product always produces a scalar (real number).

Vector Multiplication

Some notes:

- I've left out the geometric definition of the dot product, however you can learn about it in the resources listed at the end.
- There is another kind of vector multiplication, the *cross product* that I am leaving out because it is not relevent to this context.

Vector Magnitude/Norms

I mentioned that a vector represents a direction and a magnitude. The magnitude can be determined by calculating a *norm*.

There are various norms, but here we will focus on the ℓ_2, or *Euclidean* norm.

Geometrically, the ℓ_2 norm gives us the length of the straight line from the origin to the point given by the vector coordinates.

A norm is represented with two parallel lines either side of the vector:

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}.$$

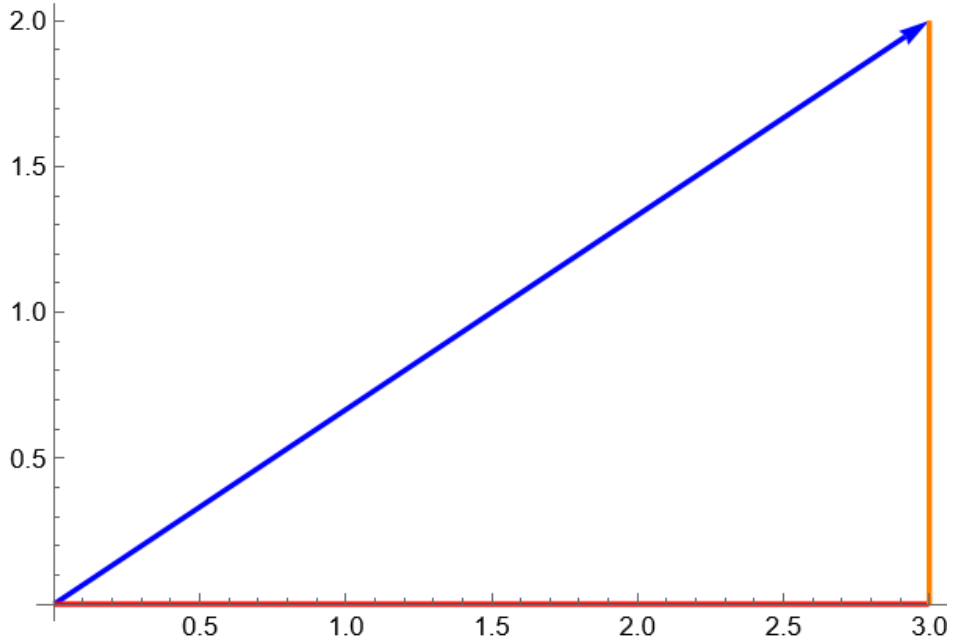
Vector Magnitude/Norms

Recall, from the Pythagorean theorem, that the length of the hypotenuse of a right-angle triangle is given by

$$c = \sqrt{a^2 + b^2}$$

where c is the hypotenuse and a and b are the two other sides.

$$\begin{aligned} \left\| \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right\|_2 &= \sqrt{3^2 + 2^2} \\ &= \sqrt{13} \\ &\approx 3.6 \end{aligned}$$



This is equivalent to the ℓ_2 norm in $\mathbb{R}^2$.

Matrices

A *matrix* is a rectangular array of numbers. For example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

is an array with 2 rows and 3 columns.

Matrix Notation

We usually name matrices with capital letters. Here is a matrix, A, with n rows and m columns, that is, $A \in \mathbb{R}^{n \times m}$:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}.$$

You will notice that denote the elements of the array with lower-case letters, and index them using the subscript, such that a_{ij} is the element of matrix A in the ith row and jth column.

Matrix Notation

In machine learning, you will often be using linear algebra in the context of *instances* and *features*, where an instance is one sample in your dataset and its features are its features are the measured values for that instance.

You will usually see the features of an instance represented as a column vector. For example:

$$x^{(i)} = \begin{bmatrix} 1 \\ 2.7 \\ -25.65 \end{bmatrix}$$

might be the ith instance of a dataset with $n = 3$ features.

We can combine all m instances of a dataset into a matrix, where each row represents one instance:

$$X = \begin{bmatrix} \left(x^{(1)}\right)^\top \\ \left(x^{(2)}\right)^\top \\ \vdots \\ \left(x^{(m)}\right)^\top \end{bmatrix} = \begin{bmatrix} \vdots & \vdots & \vdots \\ 1 & 2.7 & -25.65 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Matrix Addition

Matrix addition is performed the same way as vector addition, element-by-element. As such, we can only add two matrices of the same size.

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} + \begin{bmatrix} 4 & 2 & 0 \\ 0 & 5 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 3 \\ 2 & 9 & 14 \end{bmatrix}.$$

In general, if $C = A + B$, where $A, B \in \mathbb{R}^{n \times m}$ then:

$$C = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1m} + b_{1m} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2m} + b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & a_{n2} + b_{n2} & \cdots & a_{nm} + b_{nm} \end{bmatrix}.$$

Matrix Multiplication

Matrix multiplication is performed much like the dot product of vectors, where we treat the rows of the matrix on the left, and the columns of the matrix on the right, as the vectors.

Here is a small example:

$$\begin{aligned}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 4 & 2 \\ 0 & 5 \end{bmatrix} &= \begin{bmatrix} 1\times 4 + 2\times 0 & 1\times 2 + 2\times 5 \\ 2\times 4 + 4\times 0 & 2\times 2 + 4\times 5 \end{bmatrix} \\ &= \begin{bmatrix} 4+0 & 2+10 \\ 8+0 & 4+20 \end{bmatrix} = \begin{bmatrix} 4 & 12 \\ 8 & 24 \end{bmatrix}.\end{aligned}$$

Matrix Multiplication

Because we take the product of the rows of one matrix with the columns of the other, we must ensure that these are the same dimension.

That is, if we have a matrix $A \in \mathbb{R}^{n \times m}$ and want to find $C = AB$, we must have that $B \in \mathbb{R}^{m \times q}$ for some q.

For example, let $A \in \mathbb{R}^{2 \times 3}$ and $B \in \mathbb{R}^{3 \times 4}$:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}, B = \begin{bmatrix} 4 & 2 & 0 & 1 \\ 0 & 5 & 8 & 3 \\ 2 & 0 & 1 & 4 \end{bmatrix}.$$

We can find AB, but not BA, since B has 4 columns and A has 2 rows.

The resulting matrix, $C = AB$, where $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times q}$, will have n rows and q columns, that is, $C \in \mathbb{R}^{n \times q}$.

Matrix Transpose

The *transpose* of a matrix switches the rows and the columns, such that a $n \times m$ matrix becomes a $m \times n$ matrix:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

For example, let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

then

$$A^\top = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

The Identity Matrix

A *square matrix* is a matrix with the same number of rows and columns. Square matrices have some special properties, including the existence of an *identity matrix*.

The identity matrix has ones on the diagonal, and zeros everywhere else. For example, the 3×3 identity matrix, $I \in \mathbb{R}^{3\times 3}$ looks like this:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

If $A \in \mathbb{R}^{n\times n}$, then:

$$IA = A = AI.$$

Matrix Inverse

Another property of square matrices is that some of them have an *inverse*.

The inverse of square matrix A, given by A^{-1} has the propert that:

$$A^{-1}A = I = AA^{-1}.$$

For example, let's say we have

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

then

$$A^{-1} = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$$

Matrix Inverse

Some notes:

- We won't be going into details of how to find an inverse matrix here here, but you can learn more in the suggested readings at the end.
- The key details to note are that only square matrices have inverses, and not all square matrices are invertible.
- There are certain properties that make a matrix invertible. Again, if you want to know more, you can look to the suggested readings.

Converting Equations to Vector Form

Earlier, we saw an example of how to represent the dot product of two vectors as an equation:

$$u \cdot v = \sum_{i=1}^{n} u_i v_i$$

In machine learning, we often want to convert equations and functions to vector or matrix form so that they can be more efficiently represented and computed.

For example, in the CITS5508 lecture 3 you will see a linear regression model represented in vector form:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + ... + \theta_n x_n = \theta^\top \mathbf{x}$$

But how did we get there?

Converting Equations to Vector Form

In the case of linear equations and vectors, it is fairly straightforward.

When we have a sum from $i = 1$ to n of two lists, we can simply take the dot product.

$$\hat{y} = \theta_0 + \sum_{i=1}^{n} \theta_i x_i = \theta_0 + \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Converting Equations to Vector Form

When there is are additionial features in one list, like θ_0 in our $\hat{y}$, we need to ensure that we have two vectors of the same size, so we might add some ones:

$$\hat{y} = \theta_0 + \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

So if we let $\mathbf{x} = \begin{bmatrix} 1 & x_1 & x_2 & \dots & x_n \end{bmatrix}$ we have $\hat{y} = \theta^\top \mathbf{x}$.

Converting Equations to Matrix Form

In the CITS5508 lecture 3 you will also the the Residual Sum of Squares (RSS) represented in matrix form:

$$\text{RSS}\left(\mathbf{X}, h_\theta\right) = \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} - y_i\right)^2 = \left\|\mathbf{y} - X\theta\right\|^2,$$

where X is our matrix of instances and features, $\mathbf{x}^{(i)}$ is the column vector representing the features of the ith intance, and $\mathbf{y}$ is our vector of predicted values.

Converting Equations to Matrix Form

To get from the summation representation to the norm representation, we need to think about how we can represent the term we are summing.

Let's create a new matrix A, where each entry is one of terms to be squared in our sum. That is:

$$A_i = \theta^\top \mathbf{x}^{(i)} - y_i.$$

That is,

$$A = \begin{bmatrix} \theta^\top \mathbf{x}^{(1)} - y_1 \\ \vdots \\ \theta^\top \mathbf{x}^{(n)} - y_n \end{bmatrix}.$$

Notice that A is, in fact, an $n \times 1$ matrix, i.e. a column vector.

And, we can now write

$$\text{RSS}\left(\mathbf{X}, h_\theta\right) = \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} - y_i\right)^2 = \|A\|^2,$$

by the definition of the ℓ_2 norm.

Converting Equations to Matrix Form

Now, what is A in terms of X, θ and $\mathbf{y}$?

Recall that θ is a vector representing the weights of n features, and X is an $m \times n$ matrix representing the feature values of m instances.

$$\theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_n \end{bmatrix} \text{ and } X = \begin{bmatrix} \mathbf{x}^{(1)} \\ \vdots \\ \mathbf{x}^{(m)} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^{(1)} & \dots & \mathbf{x}_n^{(1)} \\ \vdots & \ddots & \vdots \\ \mathbf{x}_1^{(m)} & \dots & \mathbf{x}_n^{(m)} \end{bmatrix}.$$

So if we take $X\theta$, we get a vector of size m:

$$\begin{bmatrix} \mathbf{x}^{(1)} \\ \vdots \\ \mathbf{x}^{(m)} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_n \end{bmatrix} = \begin{bmatrix} \theta^\top \mathbf{x}^{(1)} \\ \vdots \\ \theta^\top \mathbf{x}^{(m)} \end{bmatrix}$$

Converting Equations to Matrix Form

Finally, by the rules of vector addition, we have:

$$X\theta - \mathbf{y} = \begin{bmatrix} \theta^\top \mathbf{x}^{(1)} \\ \vdots \\ \theta^\top \mathbf{x}^{(n)} \end{bmatrix} - \mathbf{y} = \begin{bmatrix} \theta^\top \mathbf{x}^{(1)} - y_1 \\ \vdots \\ \theta^\top \mathbf{x}^{(n)} - y_n \end{bmatrix} = A$$

So we have

$$\text{RSS}\left(\mathbf{X}, h_\theta\right) = \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} - y_i\right)^2 = \|A\|^2 = \|\mathbf{y} - X\theta\|^2$$

Converting Equations to Matrix Form

Some notes:

- This section, "Converting Equations to Matrix Form" drew heavily from the "Putting Equations into Matrix Form 1" lecture video of Kolter's Linear Algebra Review course
- I strongly suggest that you watch the second and third of the "Putting Equations into Matrix Form" videos

Resources and Further Reading

- Kolter's Linear Algebra Review course
  - "Lecture Videos" up to and including "Putting Equations into Matrix Form", or
  - "Course Notes" chapters 1 and 2, plus sections 3.1-3.10
- Géron's Linear Algebra Tutorial
  - Associated with the *Hands-On Machine Learning* textbook
  - Includes instructions on performing vector and matrix operations in Python
- Heij et al.'s *Econometric Methods with Applications in Business and Economics*
  - Section 3.1: Least Squares in Matrix Form
- Wikipedia's Glossary of Mathematical Symbols
- Academic Skills Centre online workshops:
  - Summation (Week 3)
  - You can register for this workshop to receive the slides, even if you don't attend. If you missed a workshop, or want last semester's slides in advance, please email me to request a copy at jacey.ligthart-smith@uwa.edu.au

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